

```
Retype new UNIX password: <retype password>
```

```
[livedc /]# /etc/init.d/sshd start
...key generation takes place...
```

```
* Starting sshd ... [ ok ]
```

From a remote machine, you can connect to the installation environment just like a normal Linux box:

```
[taka@durian]$ ssh root@IP_ADDRESS
```

Now we are ready to install Gentoo!

Installing Gentoo Linux

First we need to prepare the disks—by creating partitions and formatting them. Here, I'm assuming that you have a SATA disk located at `/dev/sda`, with no existing partitions. Let's use `fdisk` and list the partitions with the command `'p'`, to make sure that `sda` has no existing partitions and we can safely alter it:

```
[livedc /]# fdisk /dev/sda
```

```
Command (m for help): p
```

Device	Boot	Start	End	Blocks	Id	System
/dev/sda1		1	2952	21500375	83	Linux

Before we begin creating the partitions, we have to decide how much space we want to allocate for Gentoo, and in what manner. A default Gentoo installation will take around 2 GB of space. Portage (which will be explained later), located on `/usr/portage`, will take around 1 GB of space, and since downloaded source files will be stored there, expect it to gain 1 GB or more once we start installing various software.

For this specific set-up, we will allocate 20 GB of space for the root (`/`) partition, and 1 GB of swap space. Optionally, you can create a separate `/boot` partition to act as a rescue partition, which hosts kernel binaries and kernel configurations (`.config`); you would set up the system to unmount `/boot` after the system boot. Being able to unmount `/boot` would also reduce the risk of users modifying critical boot-loader settings.

First, let's create the root (`/`) partition on `sda1` with 20 GB of space:

```
Command (m for help): n
```

```
Command action
```

```
  e   extended
```

```
  p   primary partition (1-4)
```

```
> p
```

```
Partition number (1-4): 1
```

```
First cylinder (1-3586, default 1): <RET>
```

```
Using default value 1
```

```
Last cylinder, +cylinders or +size(K,M,G): +20G
```

```
Command (m for help): p
```

Device	Boot	Start	End	Blocks	Id	System
/dev/sda1		1	2952	21500375	83	Linux

Next, let's create a swap partition. Again, for this specific set-up we will allocate 1 GB of swap space. Traditionally, the recommended size is double the physical memory. For desktop systems with 1 GB or more of physical memory, and if you have no plans of hibernating (suspending to disk), around 1 GB of swap space is more than enough. If you want hibernation, you should set your swap space to at least the size of your physical memory. Here, we create a swap partition on `sda2` with 1 GB of space, and set the partition ID to 82, which is the ID for the Linux swap:

```
Command (m for help): n
```

```
Command action
```

```
  e   extended
```

```
  p   primary partition (1-4)
```

```
p
```

```
Partition number (1-4): 2
```

```
First cylinder (2954-3586, default 2954): <RET>
```

```
Using default value 2954
```

```
Last cylinder, +cylinders or +size(K,M,G): +1G
```

```
Using default value 3586
```

```
Command (m for help): t
```

```
Partition number (1-4): 2
```

```
Hex code (type L to list codes): 82
```

```
Changed system type of partition 2 to 82 (Linux swap / Solaris)
```

```
Command (m for help): p
```

Device	Boot	Start	End	Blocks	Id	System
/dev/sda1		1	833	3150375	83	Linux
/dev/sda2		2952	3586	1014798	82	Linux swap / Solaris

Once done, we can write the changes to the disk. There is no *undo*, so double-check everything first!

```
Command (m for help): w
```

```
The partition table has been altered!
```

```
Calling ioctl() to re-read partition table.
```

```
Syncing disks.
```

```
[livedc /]#
```

Now we create the file systems. You have a choice of various file systems; the current install CD supports ext2,